INCH-POUND

A-A-20032E
August 6, 1997
SUPERSEDING
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September 30, 1992

COMMERCIAL ITEM DESCRIPTION

POTATOES, WHITE, DEHYDRATED

The U.S. Department of Agriculture has authorized the use of this Commercial Item Description.

1. SCOPE.

1.1 This Commercial Item Description (CID) covers dehydrated, white potatoes packed in commercially acceptable containers.

2. CLASSIFICATION.

2.1 The dehydrated, white potatoes shall conform to the types, styles, and sizes, in the following list as specified in the solicitation, contract, or purchase order.

Types, styles, and sizes.

Type I - Diced, uncooked
  Style A - Without peel
  Style B - With peel
    Size 1 - 7/16 inch x 7/16 inch x 5/32 inch
    Size 2 - 3/8 inch x 3/8 inch x 1/8 inch
Type II - Mashed (precooked; rapid rehydrating)
  Style A - Granules

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any data which may improve this document should be sent to: Head, Food Quality Assurance Staff, Fruit and Vegetable Division, Agricultural Marketing Service, U.S. Department of Agriculture, STOP 0243, 1400 Independence Ave., SW, Washington, DC 20250-0243 or telephone (202) 720-9939 or FAX (202) 690-0102.

FSC 8915

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Style B - Flakes without peel
Style C - Flakes with peel
Style D - Agglomerates
Style E - Pellets
Type III - Sliced, uncooked
Style A - Without peel
Style B - With peel
Size 1 - 1/8 inch thick round random cut
Size 2 - Other
Type IV - Shredded
Style A - Uncooked
Style B - Precooked
Style C - Uncooked/Precooked
Size 1 - 7/32 inch by 7/32 inch maximum cross section, with a predominate piece length greater than ½ inch
Size 2 - Other
Type V - Other (specify)

3. SALIENT CHARACTERISTICS.

3.1 Raw ingredients: The dehydrated potatoes shall be prepared from clean, sound, white to pale yellow-fleshed potato pieces or whole potatoes.

3.1.1 Additives and optional ingredients: All additives and optional ingredients used in the preparation of the product shall be of Food Chemicals Codex purity. The optional ingredients (when required by the solicitation, contract, or purchase order) shall be as follows:

a. Milk solids (Type II Mashed)
- nonfat dry milk
- whole dried milk
- combination of nonfat dry milk and whole dried milk

b. Milk substitute (whey based products) (Type II Mashed)
c. Other (specify)

3.1.2 Preparation: The potatoes shall be dehydrated in accordance with acceptable commercial procedures, using good manufacturing practices.

3.2 DEHYDRATED PRODUCT:

3.2.1 Additives and other ingredients. The types and styles of dehydrated potatoes shall contain those additives appropriate for manufacturing.

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3.2.2 Defects: There shall not be any specific defect or combination of defects which materially affects the appearance, edibility, storage, or shipping quality of the dehydrated white potato product.

3.2.2.1 Fused product (Types I, III, and IV): More than two pieces that do not separate with slight thumbnail pressure. There shall not be more than 3.0 percent fused pieces per 100 gram sample.

3.2.2.2 Type I Diced, Style A without peel: Each individual sample unit of 100 grams of product shall contain not more than 3.0 percent, by weight, dices containing black, dark brown, or orange (scorched) discoloration(s) greater than 1/16-inch (1.6 mm) in any dimension. Peel shall be classified as a defect. No sample shall contain more than 5 defects greater than 1/4-inch (6.3 mm) in any dimension. The sample unit for evaluation is 100 grams of product.

3.2.2.3 Type I Diced, Style B with peel: Each individual sample unit of 100 grams of product shall contain not more than 3.0 percent, by weight, dices containing black or orange (scorched) discoloration(s) greater than 1/16-inch (1.6 mm) in any dimension. Peel shall not be classified as a defect. No sample shall contain more than 5 defects greater than 1/4-inch (6.3 mm) in any dimension. The sample unit for evaluation is 100 grams of product.

3.2.2.4 Type II Mashed, Style A Granules: Each sample unit shall contain not more than 80 black or dark brown specks and the average of all sample units shall not exceed 65 black or dark brown specks. The sample unit for evaluation is the surface of the product in a circle 100 mm in diameter.

3.2.2.5 Type II Mashed, Style B Flakes without peel: Each individual sample unit of 100 grams of product shall contain not more than 20 total peel, black, dark brown, or orange (scorched) specks and the average of all sample units shall not exceed 15 peel, black, dark brown, or orange (scorched) specks measuring over 1/16-inch (1.6 mm) in any dimension. Peel shall be classified as a defect. The sample unit for evaluation is 100 grams of product.

3.2.2.6 Type II Mashed, Style C Flakes with peel: Each individual sample unit of 100 grams of product shall contain not more than 20 total black or orange (scorched) specks and the average of all sample units shall not exceed 15 black or orange (scorched) specks measuring over 1/16-inch (1.6 mm) in any dimension. Numerous brown peel specks will be visible and shall not be classified as a defect. The sample shall contain no extraneous material. The sample unit for evaluation is 100 grams of product.

3.2.2.7 Type II Mashed, Style D Agglomerates and Style E Pellets: Each individual sample unit shall contain not more than 40 black, dark brown, or orange (scorched) specks, larger than a sharp pencil point dot, and the average of all sample units shall not exceed 30 black, dark brown,

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or orange (scorched) specks. The sample unit for evaluation is the surface of the product in a circle 100 mm in diameter.

3.2.2.8 Type III Sliced, Style A without peel: Each individual sample unit of 100 grams of product shall contain not more than 20 black, dark brown, or orange (scorched) discoloration(s) greater than 1/4-inch (6.3 mm) in any dimension and the average of all sample units shall not exceed 15 black, dark brown, or orange (scorched) defects(s). Peel shall be classified as a defect. The sample unit for evaluation is 100 grams of product.

3.2.2.9 Type III Sliced, Style B with peel: Each individual sample unit of 100 grams of product shall contain not more than 20 black or orange (scorched) discoloration(s) greater than 1/4-inch (6.3 mm) in any dimension and the average of all sample units shall not exceed 15 black, or orange (scorched) defects(s). Peel shall not be classified as a defect. The sample unit for evaluation is 100 grams of product.

3.2.2.10 Type IV Shredded: Each individual sample unit of 100 grams shall contain not more than 3.0 percent, by weight, shreds containing black, dark brown, or orange (scorched) discoloration(s) greater than 1/8-inch (3.2 mm) in any dimension. The sample unit for evaluation is 100 grams of product.

3.2.3 Product shape:

3.2.3.1 Type II Mashed, Style D Agglomerates: The dehydrated potatoes shall be in porous random shape agglomerated potato pieces, where by the final drying has been accomplished by air drying. The agglomerated potato pieces shall be free from clumps that will not break under slight finger pressure. Rehydration will be such that the product performs as an instant mashed potato mix after whipping by hand to form the appearance and functionally of instant mashed potatoes. In a 100 gram sample, the finished product granulation shall be of such size that none of the dehydrated potato pieces shall be retained on a U.S. Standard No. 6 sieve, that a minimum of 70 percent, by weight, shall be retained on a U.S. Standard No. 40 sieve, and a maximum of 30 percent, by weight, shall pass through a U.S. Standard No. 40 sieve. Place the sieves in a Rotap, W.S. Tyler or equivalent and shake for 30 seconds with the tapper raised.

3.2.3.2 Type II Mashed, Style E Pellets: The dehydrated potato pieces shall be in porous pelletized form, free flowing, and free from clumps that will not break under slight finger pressure. The dehydrated potato pieces final drying shall be accomplished by air drying. In a 100 gram sample, the finished product granulation shall be of such size that a minimum of 80 percent, by weight, shall be retained on a U.S. Standard No. 6 sieve and a maximum of 10 percent, by weight, shall be retained on a U.S. Standard No. 8 sieve.

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3.3 Moisture: Unless otherwise specified in the solicitation, contract, or purchase order, moisture for all types and styles of dehydrated, white potatoes shall not contain more than 9.0 percent moisture.

3.4 Reducing sugars (calculated as percent invert sugar, dry weight basis): Unless otherwise specified in the solicitation, contract, or purchase order, the reducing sugars for all types and styles of dehydrated, white potatoes shall not exceed 4.0 percent.

3.5 Sulfite content: Non-sulfited products are an option for most types of dehydrated, white potatoes. 1/ When sulfite is used, the sulfite content for all types and styles of dehydrated, white potatoes shall not exceed **500 ppm.** 2/ 3/

1/ Sulfites are naturally occurring in potatoes, and shall not exceed 10 ppm in non-sulfite added potatoes.
2/ Calculated as sulfur dioxide. ppm = parts per million
3/ Sulfite levels shall not exceed the maximum levels established under the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

3.6 REHYDRATED PRODUCT: After rehydration in accordance with label directions.

3.6.1 Flavor and odor: The rehydrated or prepared potatoes shall have a good flavor and odor typical for the type and style of potatoes. The product shall be free from objectionable or foreign flavor or odor (e.g., sour, musty, rancid, etc.).

3.6.2 Type I Diced, Type III Sliced, Type IV Shredded: The rehydrated or prepared diced, sliced, and shredded potato units must approximate the size specified and must have a mealy texture typical of prepared potatoes.

3.6.3 Type II Mashed: Following the manufacturer's label directions the rehydrated or prepared product shall have a consistency typical of mashed potatoes.

3.6.4 All types except for styles with peel: For all types except those styles with peel, the rehydrated or prepared product shall have a good, typical, cooked potato color ranging from white to pale yellow. The color shall be bright and free from gray or brown colors.

3.6.5 Styles with peel: For all styles with peel, the rehydrated or prepared product shall have a good, typical, cooked potato color ranging from white to pale yellow with peel added.

3.7 Foreign material: The dehydrated, white potatoes shall be free from extraneous matter. There shall be no foreign material such as, but not limited to, dirt, glass, wood, paint, filth, insects, metal, etc.

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3.8 Shelf life: Unless otherwise specified in the solicitation, contract, or purchase order, the dehydrated, white potatoes shall have a shelf life of at least 12 months from date of production/packaging.

3.9 Analytical requirements: When specified in the solicitation, contract, or purchase order and analyzed by United States Department of Agriculture laboratories, the dehydrated, white potatoes shall be as stated in paragraphs 3.3 to 3.5.

3.9.1 Analytical procedure: Three containers shall be selected at random.

3.9.2 Preparation of sample: The sample to be analyzed shall be a one-pound composite sample derived from a composite of the three containers. Prepare a clean, dry Waring Blendor or similar mill by grinding a sample and then discarding it. Grind approximately 100 grams of the well-mixed sample in the prepared Waring Blendor or similar mill for one minute. (Type II Mashed, Style A Granule potatoes need not be ground or screened.) Place the ground material in a small, dry, tightly-closed container for at least ½ hour. Pass the ground material over a U.S. Standard No. 20 sieve nested on a U.S. Standard No. 40 sieve. That portion which passes through the 20 mesh screen but not the 40 mesh screen is to be analyzed and shall be weighed immediately or placed in a small, dry container with a tightly closed lid until ready for analysis (a one-half pint Mason jar is satisfactory).

3.9.2.1 Further preparation of sample for reducing sugars test: Weigh 25 grams of potato granules or ground, sieved material into a 250 mL centrifuge bottle, add 75 mL of water, mix, and allow to stand for ½ hour. Add 100 mL of 95 percent ethyl alcohol plus 1 gram of calcium carbonate. Heat in a hot water (90°C) bath for 1 hour while stirring frequently. Centrifuge sample at 1,500 rpm for 10 minutes and decant the liquid portion into a 600 mL beaker.

Repeat extraction three more times, each time using 75 mL of 80 percent ethyl alcohol, heating in the hot water (90°C) bath for 15 minutes while stirring frequently. Centrifuge at 1,500 rpm for 10 minutes and decant the liquid into the 600 mL beaker. Evaporate combined extracts on a steam bath to a volume of 50 to 75 mL. Quantitatively transfer extract into 250 mL volumetric flask with hot distilled water, cool, add 10 mL of saturated, neutralized lead acetate 1/, and bring to volume. Filter liquid through No. 2V Whatman or equivalent filter paper into a 250 mL Erlenmeyer flask containing about 4 grams of potassium oxalate. Mix filtrate thoroughly and allow precipitate to settle. To remove excess lead, add a few potassium oxalate crystals. Decant liquid through No. 2V Whatman or equivalent filter paper into a 250 mL glass-stoppered Erlenmeyer flask. Reagents used shall be American Chemical Society Reagent Grade.

1/ A saturated solution of neutralized lead acetate is prepared by adjusting the solution to pH 6.8 with 1N sodium hydroxide.

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3.9.2.2 Examination for moisture: Weigh to the nearest 0.1 mg, two, approximately 2-gram, samples of the ground, screened material (that portion which has passed through the 20 mesh screen and has been collected on the 40 mesh screen) into tared, dry, aluminum weighing dishes approximately 2 or 2-1/2-inches (50.8 or 63.5 mm) in diameter and 3/4-inch (19.1 mm) in depth with tight-fitting covers. Place the dishes with cocked lids in a vacuum-oven and dry for 6 hours at 70°C under a pressure of not more than 100 mm of mercury. During drying, admit to the oven a slow current of air (approximately 2 bubbles per second), dried by passing through concentrated sulfuric acid. Remove the dishes, place the lids on tightly, and allow them to cool in a desiccator before weighing. Calculate the percent of moisture in the original samples and report the average of the two samples.

3.9.3 Analytical testing: The sample shall be analyzed using the following Official Methods of Analysis of the AOAC International methods or other methods as follows:

Test	Method
Moisture	934.06 (first paragraph) and paragraph 3.9.2.2 of this CID.
Reducing Sugars	906.03 and paragraph 3.9.2.1 of this CID.
Sulfite Content	990.28

3.9.4 Test results: The test results for moisture and reducing sugars shall be reported to the nearest 0.1 percent. Test results for sulfite content shall be reported to the nearest 10 ppm. Any result not conforming to the finished product requirements shall be cause for rejection of the lot.

4. REGULATORY REQUIREMENTS.

4.1 The delivered dehydrated, white potatoes shall comply with all applicable Federal and State mandatory requirements and regulations relating to the preparation, packaging, labeling, storage, distribution, and sale of the dehydrated, white potatoes within the commercial marketplace. Delivered dehydrated, white potatoes shall comply with all applicable provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

5. QUALITY ASSURANCE PROVISIONS.

5.1 Product conformance. The dehydrated, white potatoes provided shall meet the salient characteristics of this CID, conform to the producer's own specifications, standards, and quality assurance practices, and be the same dehydrated, white potatoes offered for sale in the commercial market. The Government reserves the right to require proof of such conformance.

5.2 Quality assurance. When required in the solicitation, contract, or purchase order that product quality or acceptability or both be determined, the Processed Products Branch (PPB),

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Fruit and Vegetable Division, Agricultural Marketing Service, U.S. Department of Agriculture, shall be the certifying activity and shall make the determination in accordance with applicable PPB procedures. The dehydrated, white potatoes shall be examined or analyzed or both in accordance with applicable provisions in this CID, solicitation, contract, or purchase order, and, when applicable, the United States Standards for Condition of Food Containers in effect on the date of solicitation.

6. PACKAGING.

6.1 Preservation, packaging, packing, labeling, and case marking. Preservation, packaging, packing, labeling, and case marking shall be as specified in the solicitation, contract, or purchase order.

7. NOTES.

7.1 Purchasers shall specify:

- Type(s), style(s) and size(s), as appropriate.
- Optional ingredients, as appropriate.
- Moisture content, if other than a maximum of 9.0 percent.
- Reducing sugar content, if other than a maximum of 4.0 percent.
- For Type II Mashed, Styles B and C Flakes:
  - With or without sulfiting agents
- For Type II Mashed, Style D Agglomerates:
  - With or without lumps

7.2 Sources of documents.

7.2.1 Sources of information for nongovernmental document are as follows:

Copies of the Official Methods of Analysis of the AOAC International may be obtained from: **AOAC International, 481 North Frederick Avenue, Suite 500, Gaithersburg, MD 20877.**

Copies of the Food Chemicals Codex may be purchased from: **National Academy Press, 2101 Constitution Avenue, NW, Washington, DC 20418.**

Copies of the Reagent Chemicals may be obtained from: **Sales and Distribution Department, American Chemical Society, 1155 16th Street, NW, Washington, DC 20036.**

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7.2.2 Sources of information for governmental documents are as follows:

Applicable provisions of the Federal Food, Drug, and Cosmetic Act are contained in 21 CFR Parts 1 to 199. This document may be purchased from: **Superintendent of Documents, ATTN: New Orders, P.O. Box 371954, Pittsburgh, PA 15250-7954.** Credit card (MasterCard or Visa) purchases may be made by calling the Superintendent of Documents on (202) 512-1800.

Copies of the United States Standards for Condition of Food Containers are available from: **Chairperson, Condition of Container Committee, STOP 0243, 1400 Independence Avenue, SW, Washington, DC 20250-0243.**

Civil agencies and other interested parties may obtain copies of this CID from: **General Services Administration, Federal Supply Service, Specifications Section, Suite 8100, 470 East L'Enfant Plaza, SW, Washington, DC 20407.**

Military activities should submit requests for copies of this CID to: **Standardization Documents Order Desk, Defense Automation Printing Service, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.**

MILITARY INTERESTS:

Military Coordinating Activity

Army - GL

Custodians

Army - GL
Navy - SA
Air Force - 35

Review Activities

Army - MD, QM
Navy - MC
DLA - SS

CIVIL AGENCY COORDINATING ACTIVITIES:

DOJ - BOP
HHS - NIH, IHS
USDA - FV
VA - OSS

PREPARING ACTIVITY:

USDA - FV

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